

Identifying Brain-Wide Organization of Stimulus and Choice Signals in Mouse Visual Decision-Making

Impact Scholars Program Project Proposal



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neuromatch

Background & Significance



Using the IBL Dataset



How are sensory and choice signals organized across the brain?



We have anatomical atlases yet no functional atlas of stimulus and choice encoding!

Why this matters?



Signal transformation Insight

- Hierarchical
- Specialized
- Intermingled



Clinical Implications

- ADHD
- Schizophrenia
- Neglect

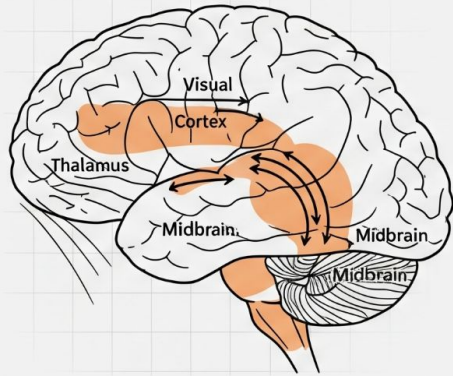


Computational Implications

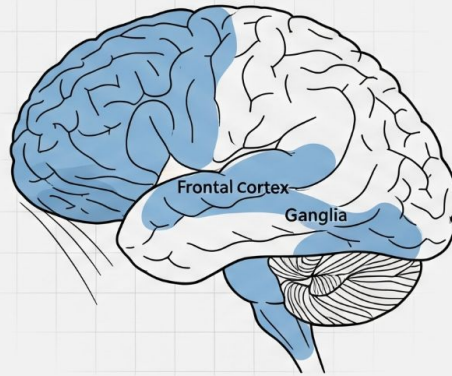
- Benchmark brainwide dataset for neural decoding
- Reproducible pipeline



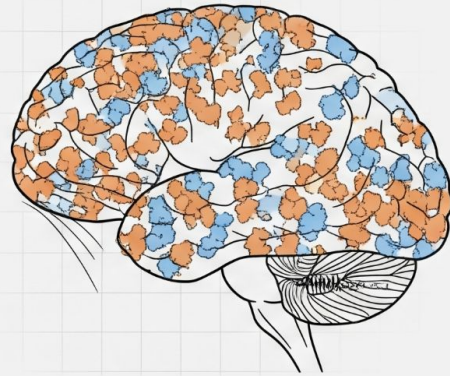
Research Question & Hypotheses



H1: Stimulus Encoding

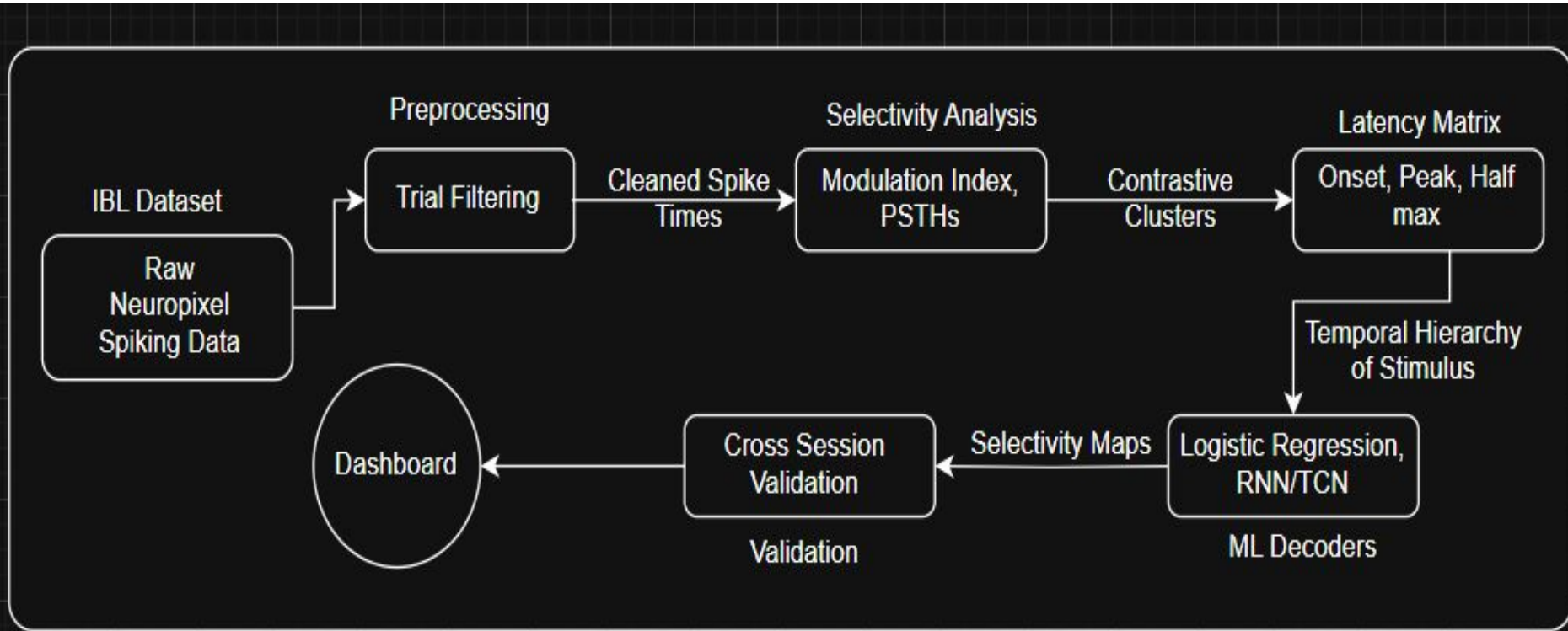


H2: Choice Encoding



H0: Widespread Integration

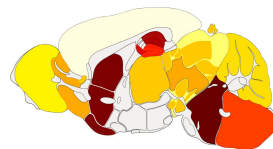
Methods Pipeline



Expected Outcomes & Impact

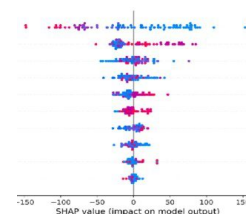
01	Scientific	First <i>functional atlas</i> of stimulus & choice encoding	
02	Applied	Inform models of decision-making deficits	
03	Computational	Reproducible AI/neuro pipeline (GLM + ML decoders)	
04	Community	Open-source tools & atlas dashboard for future work	

Atlas



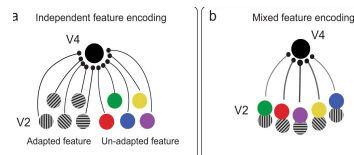
Brain regions ranked by selectivity & latency

Decoders



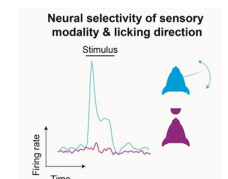
Accuracy $\geq 70\%$, feature importance

Hypothesis Test



Distinguish between specialized vs intermingled encoding

Dashboard



Interactive UI for querying selectivity/latency hierarchies



References

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- Icons used in this presentation were obtained from <https://www.flaticon.com/>

